

- 1) No. of defective hard disks } counting
- 2) No. of defective per hard disk } counting
- 3) failure time of hard disks } measuring
- 4) time of complete (a task) } measuring

Random variable are denoted by cap

X, Y, Z, U, V, W

value of Random variable :-

x, y, z, u, v, w

eg $X = \frac{1}{2}$
RV value

$$P(X=x) = P(x)$$

Discrete r.v. continuous r.v.
integer real no.

let X is a discrete r.v possible value

x	$P(X=x)$
x_1	$P(X=x_1) = P(x_1)$
x_2	$P(x_2)$
x_n	$P(x_n)$

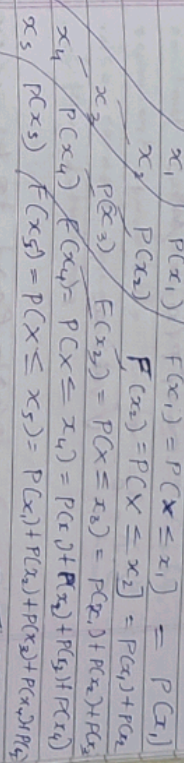
prob distribution of r.v. $X \bullet P(X) \geq 0 \forall x$

• Total prob = 1

$$\sum P(x) = 1$$

Cumulative distribution of X ($F(x)$)

$$F(x) = P(X \leq x)$$



Graph of prob distribution

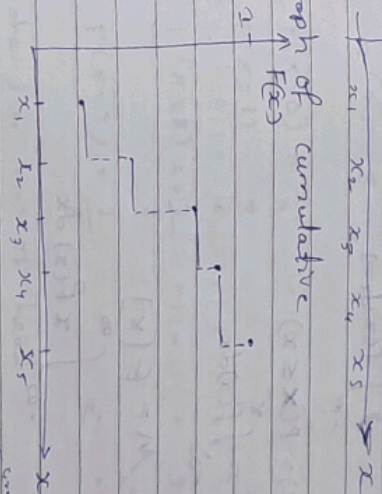
$y = P(x)$

Prob histogram

Graph of cumulative

$y = F(x)$

cumulative distribution step function



mean

$$\bar{x} = \frac{\sum x_i \cdot f_i}{N} = \frac{\sum x_i \cdot P(x_i)}{1}$$

$$\text{Mean} = M = E[X] = \sum_{i=1}^n x_i \cdot P(x_i)$$

$$\text{Variance } \sigma^2 = \frac{\sum (x_i - \bar{x})^2 \cdot f_i}{N} = \frac{\sum (x_i - \bar{x})^2 \cdot P(x_i)}{1}$$

$$\text{Variance} = \sigma^2 = E[(x_i - M)^2] = \sum_{i=1}^n (x_i - M)^2 \cdot P(x_i)$$

graph of $f(x)$ is smooth curve graph of $f(x)$ is non decreasing

For numerical computations
 $\sigma^2 = E[X^2] - \{E[X]\}^2$

$= \sum x_i^2 P(x_i) - M^2$

S.D = $\sigma = \sqrt{\text{variance}}$

$P(X=x) = P(x)$ or $f(x) \rightarrow$ Prob mass fn (P.M.F.)
 Prob. density fn (P.D.F.)

$F(x) = P(X \leq x)$

of X is a count. r.v.
 $f(x)$ is a valid {P.D.F.} if $f(x) \geq 0 \forall x$.

$\int_{-\infty}^{\infty} f(x) dx = 1$

$F(x) = P(X \leq x)$

$= \int_{-\infty}^x f(x) dx$

mean = $M = E[X]$

$= \int_{-\infty}^{\infty} x f(x) dx$

$\text{Var} = \sigma^2 = E[(X-M)^2]$

$= \int_{-\infty}^{\infty} (x-M)^2 f(x) dx$

$\sigma^2 = E[X^2] - \{E[X]\}^2$

$= \int_{-\infty}^{\infty} x^2 f(x) dx - M^2$

- 3 coins are tossed. It's a binomial distribution.
- Find the probability distribution of no. of heads obtained.

construct cumulative distribution function, represent probability functions & CDF area also compute mean (Average no. of heads) & variance of the distribution

$\Rightarrow X =$ no. of heads obtained

Possible values: 0, 1, 2, 3

X is discrete r.v.

The sample space

$S = \{ (H,H,H), (H,H,T), (H,T,H), (H,T,T), (T,H,H), (T,H,T), (T,T,H), (T,T,T) \}$

$n(S) = 8$

$P(X=0) = \frac{1}{8}$

$P(X=1) = \frac{3}{8}$

$P(X=2) = \frac{3}{8}$

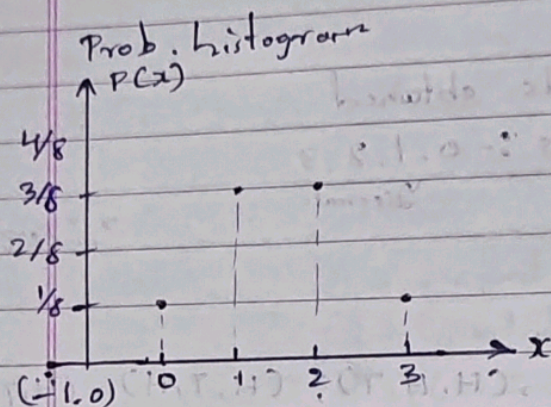
$P(X=3) = \frac{1}{8}$

Probab. distr. no. of head

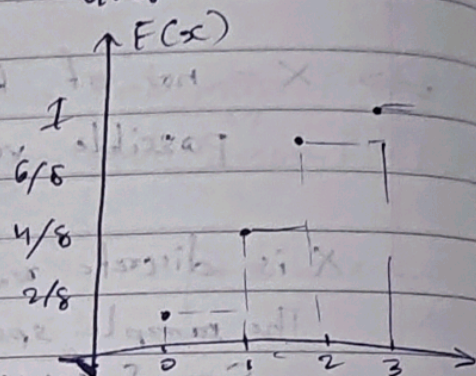
x	$P(X=x)$
0	$\frac{1}{8}$
1	$\frac{3}{8}$
2	$\frac{3}{8}$
3	$\frac{1}{8}$

C d F

x	$F(x) = P(X \leq x)$
0	$\frac{1}{8}$
1	$\frac{4}{8}$
2	$\frac{7}{8}$
3	1 or $\frac{8}{8}$



Graph of $F(x)$



$$\text{Mean} = \mu = E(X)$$

$$= \sum_{x=0}^3 x \cdot P(x)$$

$$= 0 \cdot P(0) + 1 \cdot P(1) + 2 \cdot P(2) + 3 \cdot P(3)$$

$$= \left(0 \cdot \frac{1}{8}\right) + \left(1 \cdot \frac{3}{8}\right) + \left(2 \cdot \frac{3}{8}\right) + \left(3 \cdot \frac{1}{8}\right)$$

$$= \frac{12}{8}$$

$$= 1.5$$

$$\text{Variance} = \sigma^2 = E[X^2] - \mu^2$$

$$\text{Now } E[X^2] = \sum_{x=0}^3 x^2 \cdot P(x)$$

$$= (0^2 \cdot P(0)) + (1^2 \cdot P(1)) + (2^2 \cdot P(2)) + (3^2 \cdot P(3))$$

$$= \frac{1}{8} \left[0^2 \cdot \frac{1}{8} + 1^2 \cdot \frac{3}{8} + 2^2 \cdot \frac{3}{8} + 3^2 \cdot \frac{1}{8} \right]$$

$$= 0 + \frac{3}{8} + \frac{12}{8} + \frac{9}{8} = \frac{24}{8} = 3$$

$$\therefore \text{variance} = 3 - (1.5)^2 = 3 - 2.25 = 0.75$$

$$\text{S.D } \sqrt{\text{var}} = \sqrt{0.75} = 0.866$$

3. X takes integer values
 X is a discrete random variable

i) $k = ?$

We know that Total probability = 1

i.e.

$$\sum p(x) = 1$$

$$\frac{1}{8} + \frac{2}{8} + k + \frac{2}{8} + \frac{1}{8} = 1$$

$$\frac{6}{8} + k = 1$$

$$F \text{ i.e. } \frac{3}{4} + k = 1$$

$$\therefore k = 1 - \frac{3}{4} = \frac{1}{4}$$

$$(1) \frac{1}{8} + (0) \frac{2}{8} + (1) \frac{1}{4} + (1) \frac{1}{8}$$

$$\frac{1}{8} + \frac{2}{8} + \frac{2}{8} + \frac{1}{8} = 1$$

x	$P(x)$	$F(x)$
-2	$\frac{1}{8}$	$\frac{1}{8}$
-1	$\frac{2}{8}$	$\frac{3}{8}$
0	$\frac{1}{4}$	$\frac{5}{8}$
1	$\frac{2}{8}$	$\frac{7}{8}$
2	$\frac{1}{8}$	1
	1	

$$\text{Mean} = E[X]$$

$$\sum_{x=-2}^2 x \cdot p(x) = \frac{1}{8}$$

$$\text{Variance} = \sigma^2 = E[X^2] - M^2$$

$$\sum_{x=-2}^2 x^2 \cdot p(x)$$

$$= 1.5 - \left[\frac{1}{8}\right]^2 = 1.484$$

Q. $P(X < 1) = P(-2) + P(-1) + P(0)$

$$= \frac{1}{8} + \frac{2}{8} + \frac{1}{4}$$

$$= \frac{5}{8}$$

$$\frac{1}{8} + \frac{2}{8} + \frac{2}{8} + \frac{1}{8}$$

$$\frac{12}{8} = \frac{3}{2} = 1.5$$

$$\begin{aligned}
 P(X \leq 1) &= P(-2) + P(-1) + P(0) + P(1) \\
 &= \frac{1}{8} + \frac{2}{8} + \frac{1}{4} + \frac{2}{8} \\
 &= \frac{1}{8} + \frac{2}{8} + \frac{2}{8} + \frac{2}{8} \\
 &= \frac{7}{8} //
 \end{aligned}$$

$$\begin{aligned}
 P(X > 0) &= P(1) + P(2) \\
 &= \frac{2}{8} + \frac{1}{8} = \frac{3}{8} //
 \end{aligned}$$

$$\begin{aligned}
 P(X \geq 0) &= P(0) + P(1) + P(2) \\
 &= \frac{1}{4} + \frac{2}{8} + \frac{1}{8} = \dots \\
 &= \frac{5}{8} //
 \end{aligned}$$

$$\begin{aligned}
 P(-2 < X < 1) &= P(-1) + P(0) \\
 &= \frac{2}{8} + \frac{1}{4} \\
 &= \frac{4}{8} = \frac{1}{2} //
 \end{aligned}$$

$$\begin{aligned}
 P(-2 \leq X \leq 1) &= P(-2) + P(-1) + P(0) + P(1) \\
 &= \frac{1}{8} + \frac{2}{8} + \frac{1}{4} + \frac{2}{8} = \frac{7}{8} //
 \end{aligned}$$

$$\begin{aligned}
 P(-1 \leq X \leq 2) &= P(-1) + P(0) + P(1) \\
 &= \frac{2}{8} + \frac{1}{4} + \frac{2}{8} = \frac{6}{8} = \frac{3}{4} //
 \end{aligned}$$

$X \sim B(n, p)$
 Binomial Distribution:-

- Fixed no. of trials (n) - finite no.
- observing x success among n
- fixed prob of success (p)
- Discrete prob. distr.

P.D.F

$$P(X=x) = \binom{n}{x} p^x (1-p)^{n-x} \cdot \binom{n}{x} p^x q^{n-x}$$

$$x = 0, 1, 2, \dots, n$$

$$q = 1 - p$$

$$0 < p < 1$$

Symbolically
 $X \sim B(n, p)$

Parameters

ex: $n = 5, p = \frac{1}{3}$

$$P(X=x) = \binom{5}{x} \left(\frac{1}{3}\right)^x \left(\frac{2}{3}\right)^{5-x}$$

$$\sum_{x=0}^5 P(X=x) = 1$$

C.D.F

$$F(x) = P(X \leq x) = \sum_{y=0}^x P(y)$$

$$= \sum_{y=0}^x \binom{n}{y} p^y q^{n-y}$$

mean $= np$

variance $= npq$

In Binomial distribution
 mean $\rightarrow$ variance

For $p < \frac{1}{2}$ positively skewed

$p > \frac{1}{2}$ negatively skewed

$p = \frac{1}{2}$ symmetric

SUCCESS (defective)

10 items $\rightarrow$ no. of defectives

defective $\rightarrow x$

contains defects

how many?

no. of defect. per item

$p \rightarrow 0$ (small) $p \rightarrow 1$ (large)

if p is $\rightarrow 0$ (small) p is large

if p is $\rightarrow 1$ (large) p is small

if p is $\rightarrow 0.5$ (constant) p is avg

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bomb drop

strike (hit)	do not strike (no hit)		
success	miss		
	failure		

$p =$ a bomb will strike the target

$$p = \frac{1}{2}$$

$$q = \frac{1}{2}$$

$n =$ no. of bombs dropped

$X =$ no. of bombs striking the target

$$P(X \geq 2) \geq 0.99$$

$$n = ?$$

$$\therefore X \sim B(n, p)$$

$$P(X) = \binom{n}{x} p^x q^{n-x}$$

$$= \binom{n}{x} \left(\frac{1}{2}\right)^x \left(\frac{1}{2}\right)^{n-x}$$

$$P(X) = \binom{n}{x} \left(\frac{1}{2}\right)^n$$

Consider

$$P(X \geq 2) \geq 0.99$$

$$\text{ie } 1 - P(X \leq 2) \geq 0.99$$

$$1 - 0.99 \geq P(X \leq 2)$$

$$\therefore P(X \leq 2) \leq 0.01$$

$$P(0) + P(1) \leq 0.01$$

$$\binom{n}{0} \left(\frac{1}{2}\right)^n + \binom{n}{1} \left(\frac{1}{2}\right)^n \leq 0.01$$

$$\left(\frac{1}{2}\right)^n + n \left(\frac{1}{2}\right)^n \leq 0.01$$

$$\left(\frac{1}{2}\right)^n (n+1) \leq 0.01$$

$$\frac{1}{2^n} (n+1) \leq 0.01$$

$$n+1 \leq 2^n (0.01)$$

$$\text{or } 2^n (0.01) \geq n+1$$

minimum 11 bombs are to be dropped

$\therefore n + 11 = 20$

n	$P(X=n)$	n	$P(X=n)$
1	0.04	11	0.04
2	0.08	12	0.08
3	0.16	13	0.16
4	0.32	14	0.32
5	0.64	15	0.64
6	1.28	16	1.28
7	2.56	17	2.56
8	5.12	18	5.12
9	10.24	19	10.24
10	20.48	20	20.48

$P(X \geq 11) = 1 - P(X \leq 10)$

$n = \text{no of coins} = 8$

$N = 256$

$X = \text{no of heads}$
 $P(X) = \binom{8}{x} p^x q^{8-x}$
 $p = 0.12, q = 0.88$

$P(X) = \binom{8}{x} p^x q^{8-x}$

To estimate P no. of heads

We know that in binomial distribution

Mean = np

$\sum f_i x_i = np$

$(0 \times 2) + (1 \times 6) + (2 \times 24) + \dots + (8 \times 1)$

$= 4.074 = np$

$P = \frac{4.074}{8}$

$P = 0.509$

$q = 1 - p$
 $= 1 - 0.509$
 $q = 0.491$

$\therefore P(X) = \binom{8}{x} (0.509)^x (0.491)^{8-x}$

Theoretical Frequencies are given by

$T_x = N \cdot P(x)$
 $= 256 \times \binom{8}{x} (0.509)^x (0.491)^{8-x}$

x	T_x	T_x
0	1.00	0.865
1	8.01	7.172
2	28	26.021
3	56	53.944
4	70	69.909
5	56	57.978
6	28	30.051
7	8	8.904
8	1	1.153

$\sum T_x = 256.001$

$(0.509)^x (0.491)^{8-x}$